

Study Data Reviewer's Guide (SDRG)

Field	Value
Document	Study Data Reviewer's Guide (SDRG)
Study	TROPIC / EFC6193 / NCT00417079
Compound	Cabazitaxel (CbzP) vs Mitoxantrone (MP)
Standard (package layer)	CDISC SDTMIG **v3.4** + CDISC/NCI CT 2026-03-27 (uplifted; §5)
Standard (pristine source)	CDISC SDTMIG **v3.1.1** (PDS 2013; local `01_source_data/real_sdtm/`)
Document version	1.1 (Path A hardened)
Effective	2026-07-09
Supersedes	SDRG narrative drafts prior to `v0.1.0-demo-rc.1` Path A freeze
Product claim	**Path A only** -- controlled non-submission demonstration

0. What this package is / is not (read first)

This package **is**	This package **is not**
Study-data explanation for a **submission-style** Module 5 tree	An FDA filing / real application sequence
Real **MP-only** de-identified SDTM (N=371) as source for dual-language ADaM	Two-arm real patient-level IPD in git
Uplifted SDTMIG 3.4 **package layer** + define co-located under `m5/.../tabulations/sdtm/`	Proof that commercial P21 cleared every residual
CORE SDTM 3.4 run with **classified residuals** (F-015)	"Full CORE clean" / zero-finding claim
Week-offset / partial-date source honesty (F-017)	Day-level AE timing precision that the source never had
Path A demo with patient XPT **not redistributed**	Part 11 validated system

Binding claim: docs/PRODUCT_CLAIM.md

Source intake pack (WS-1): docs/workstreams/WS1_SOURCE_INTAKE_PACK.md

Residual risks: docs/workstreams/WS5_KNOWN_DIFFERENCES_MEMO.md

External validation slots: docs/workstreams/WS3_EXTERNAL_VALIDATION_EVIDENCE_INDEX.md

Sealed demo RC: tag v0.1.0-demo-rc.1 docs/RELEASE_NOTE_v0.1.0-demo-rc.1.md python3 scripts/verify_release.py

Review package face: 08_submission_package/m5/datasets/tropic/tabulations/sdtm/

Analysis narrative: ADRG.md

Findings disposition: 06_qc_evidence/audit/FINDINGS_DISPOSITION_BOARD.md

*****Redistribution:**** Real `*.sas7bdat`` and package `*.xpt`` are ****not**** in git (data rights + portfolio surface policy). Structure, define, SDRG/ADRG, and programs are.*

0A. CRF grounding (principal source-fidelity rule)

CRF PDF (source): 01_source_data/Sanofi CRF Tropic.pdf (EFC6193, FINAL 21-Nov-2006)

In package: blankcrf.pdf under tabulations/sdtm (source CRF copy -- not claimed as a complete annotated aCRF with page-level define origins)

Decision record: docs/workstreams/reviews/WS1_CRF_GROUNDING_D012_2026-07-09.md

Rule

Do	Do not
State what the **form collected**	Imply "trial never collected X" when the CRF has a field for X
State what **PDS retains** (and at what precision)	Blame study design for public-extract reductions
Split gaps: CRF non-collection vs extract reduction vs programming scope	Invent values for null/stripped fields

Domain grounding summary (audit D-012)

Domain	CRF form (examples)	What sponsor collected	What PDS extract retains	Limitation class
AE	Adverse Event Form `O.1_AE_1`	Diagnosis; status; grade; relationship to study treatment; action taken; corrective therapy; outcome (incl. fatal); seriousness Y/N + seriousness criteria	Terms/MedDRA; `AETOXGR` (incl. grade 5 fatal); `AEREL`/`AEACN`/`AECONTRT`/`AEOUT` largely populated; `AESER` + criteria flags largely present	**Class A** (core safety retained). **Class B** for **date precision** (CRF day/month/year ? PDS week offsets `AESTWK`/`AEENWK`). Residual blank `AESER` on a minority of rows = null/incomplete, not "column never collected"
LB	Hematology `LABH_1`, Biochemistry `LABB_1`, PSA	CBC panel; electrolytes (Na/K/Cl/bicarb); LFTs; creat/BUN; glucose; testosterone; PSA	Matching `LBTESTCD` set including `SODIUM`, `K`, `CL`, `BI CARB`, `ALP`, `CREAT`, `PSA`, etc.	**Class A** -- electrolytes are **on CRF and in extract** (not a TROPIC "electrolyte never collected" story)
ECOG	VS panel ECOG 0-4	Performance status 0-4	`VSTESTCD`=ECOG with observed 0-4	**Class A**
DS	End of Treatment `O.ENDTT_2`; End of Study `O.ENDST_1`	Main reason for stopping treatment/study (progression, AE, completed, LTFU, subject request, death, other, ...)	`DSCAT`/`DSSCAT`/`DSDECOD`/`DSTERM` carry those concepts	**Class A** for reason concepts present in extract

Explicit non-claims: full aCRF annotation package; day-true AE onset dates in public extract; commercial filing CRF provenance.

Package domain scope (analysis-scoped SDTM)

Layer	Content
Pristine PDS (local)	34 `*.sas7bdat` domains (incl. EG, MH, PE, SV, IE, CD, CX, trial design fragments, etc.)
Package / `define_sdtm.xml`	**18** datasets: DM, EX, AE, LB, CM, DS, VS, LS, PN + SUPPAE/CM/DM/DS/EX/LB/LS + **TA, TS**

Domains present in PDS but not packaged (by design for this Path A analysis package):
CD, CX, EG, IE, MH, PE, PR, SC, SV, TE, TI, TV (+ related SUPP).
Do not claim the Module 5 SDTM folder is a full copy of every PDS domain.

Full E2E audit: docs/workstreams/reviews/WS1_SDTM_E2E_AUDIT_2026-07-09.md

Material extract residuals (SDTM E2E)

ID	Finding	Honesty
F-028	One subject has `EXTRT=XRP6258` (10 rows) while **all** DM rows are `ARM=MITOXANTRONE/PREDNISONE`	Arm authority = **DM/ADSL**, not EXTRT alone; do not silently re-code EX
AE coverage	AE subjects **357/371** (14 subjects with no AE rows)	Any-AE rates must use ADSL denominator
F-026	~1134 BASELINE AE skeleton rows (blank AESER)	TEAE analyses use `TRTEMFL`

1. Source data normalization & integrity controls

Source data are the official Sanofi de-identified SDTM for NCT00417079 (2013), accessed via Project Data Sphere (PDS) as SAS .sas7bdat. Provenance: root README 00_governance/REPRODUCIBILITY.md.

Staging: L_staging_ingest.sas (and R twin v_staging_ingest.R) ingests realsdtm.<domain>, transpose-merges SUPP--, and coerces character continuous indicators to numeric where required.

***Single-arm source limitation (Path A critical):** PDS public data here is **MP only (N=371)**. Cabazitaxel is **not** present as source SDTM. Dual-language ADaM recon is MP-only. Synthetic/reconstructed CbzP is merged only at TFL reporting ([`tfl\_generation.R`](../05_outputs/tfl/tfl_generation.R)) -- non-confirmatory (F-003). Do not describe this as a clean two-arm double-programming of trial IPD.*

Database write-protection architecture

* `realsdtm` libref ? `01\_source\_data/real\_sdtm` with `access=readonly` in
[`00\_config.sas`](../04_analysis_datasets/programs/sas/00_config.sas)

- * `staging` libref writable for SUPP-merged intermediates during ODA runs
 - * Mapped intermediates redirect under `04\_analysis\_datasets/adam/` (local build outputs; not the git portfolio face)
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2. SDTM Domain Mapping Summary

Standard SDTM mapping structures were built in `S_sdtm_mapping.sas` from trial-era SDTM-IG 3.1.1 source data; SAP v4.0 treats this as source provenance and requires the release package to use the governed SDTMIG 3.4 uplift layer:

- * **DM (Demographics):** Unique subject identifier `USUBJID` constructed via `STUDYID || '-' || SITEID || '-' || SUBJID`. Randomization date `RANDDT` and treatment start date `TRTSDT` mapped to standard ISO 8601 date fields.
 - * **EX (Exposure):** Normalised cycle-level actual administered doses (`EXDOSE` in mg).
 - * **AE (Adverse Events):** Coded utilizing MedDRA dictionaries into `AEDECOD`, `AEBODSYS`, and CTCAE-style grades (`AETOXGR`). **CRF grounding (D-012):** the Adverse Event form collected seriousness, relationship to study treatment, action taken, outcome (incl. fatal), and seriousness criteria -- and those concepts are **largely present** in the PDS extract (`AESER`, `AEREL`, `AEACN`, `AEOUT`, `AESxxx` flags). Do **not** narrate "trial never collected seriousness." **Date Precision Note (Class B reduction):** CRF collected calendar day/month/year; the public extract carries week-offset integers (`AESTWK`, `AEENWK`). Dates are reconstructed as `RFSTDTC + (AESTWK - 1) \* 7` (± 3.5 days). That precision limit is an **extract design / de-identification property**, not a programming artefact. ADAE and TTSAE inherit it.
 - * **LB (Laboratory):** Mapped continuous Absolute Neutrophil Count (ANC) and Prostate Specific Antigen (PSA) measurements.
 - * **DS (Disposition):** Captured study completion reasons, trial exits, and survival follow-up records.
 - * **RS (Response / Efficacy Fallback):** Derived from `DS` domain where `DSDECOD` indicates progression or death. Death records are mapped to standard RS structures with `RSSTRESC = 'DEATH'` to capture survival outcomes cleanly as efficacy checkpoints.
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3. Reference Ranges & Baseline Criteria

- * Baseline lab and vitals measurements are defined as the last non-missing assessment completed prior to first exposure (`ADY <= 0`).
 - * Normal reference ranges (`LBNRLO` and `LBNRHI`) were preserved from raw PDS metadata. Lab values outside these ranges are flagged accordingly in `LBNRIND`.
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4. Known Data Limitations & Derivation Decisions

4.1 Baseline Laboratory Imputation

For subjects with missing baseline laboratory values (PSABL, ALPBL, HGBBL), population-median proxy values have been imputed in ADSL:

- * `PSABL` default: 110.0 ng/mL
- * `ALPBL` default: 140.0 U/L
- * `HGBBL` default: 11.5 g/dL
- * `ALBBL` **missing** (not collected; no imputation)

* `LDHBL` ****missing**** (not collected; no imputation)

This imputation strategy is retained as a documented implementation limitation under SAP v4.0 §14 / §18. These imputed baseline laboratory constants are schema placeholders and are NOT used as covariates or stratification factors in any efficacy model, consistent with ADRG §5.1. The primary and secondary Cox / log-rank analyses stratify only on the protocol randomization strata (pooled ECOGBL 0-1 vs 2 and MEASDISF; see 05_outputs/tfl/tfl_generation.R, compute_tte_stats() ? strata(ECOGBLGRP, MEASDISF)). ALBBL and LDHBL in particular are missing for all subjects (no subject-level source available; both tracks leave them missing with blank flags), so they carry no subject-level information; they are retained purely to satisfy the ADaM schema and should be read as "not available," not as analysis inputs.

4.2 Supplemental Domain Ingestion

Domains LS (Lesion) and PN (Pain/Numeric) do not have supplemental (SUPPLS, SUPPPN) datasets in the PDS source data. The %transpose_supp() macro gracefully handles this via the supp_exists = 0 guard path, copying the primary domain directly without SUPP merge.

4.3 Country and Region Assignment

The DM domain in the source data does not contain country-of-study-site information. COUNTRY and REGION are not present in the de-identified release and are not derived (see the note in B_bimo_generation.sas); no placeholder geography is assigned. Geographic subgroup analyses are not part of SAP v4.0 reporting and are not reported.

4.4 Hardcoded Demographic Constant (SEX = 'M')

The demographics domain (DM) contains a hardcoded variable SEX = 'M' assigned to all subjects in A_adsl_generation.sas. This is a clinical decision consistent with the trial protocol for metastatic castration-resistant prostate cancer (mCRPC), which is an exclusively male patient population. To ensure metadata conformity, the Define-XML codelist references are maintained; however, no female subjects are present in the analysis dataset.

4.5 Partial/Imprecise Source Date Values (CM, LB, LS, PN)

The independent R SDTM validation (04_analysis_datasets/programs/r/v_sdtm_validation.log) raises four [WARNING]s flagging partial or imprecise ISO-8601 date values in source date fields: CMSTDTC (CM), LBDTC (LB), LSDTC (LS), and PNDTC (PN) -- e.g. ----07, --12-26, 2009---04. These are expected manifestations of the source PDS public-release date precision (the same root cause documented in §2 and §5.1), not programming defects: the values are carried through as-is rather than fabricating spurious precision. They surface as WARNINGS (not ERRORS); no analysis depends on day-level precision in these fields. No action required.

5. SDTMIG 3.4 Conformance Uplift (2026-06-20)

The pristine source SDTM (01_source_data/real_sdtm/, PDS 2013) was authored to SDTMIG 3.1.1, which is below the FDA Data Standards Catalog support floor. A derived, conformance-uplifted SDTM layer was produced to SDTMIG 3.4 + CDISC/NCI CT 2026-03-27 and is the version described by define_sdtm.xml and packaged in 08_submission_package/m5/.../tabulations/sdtm/. The raw source is never modified; the uplift is a deterministic derivation step (platform/uplift_sdtm_34.R for data, 03_metadata/define/uplift_define_34.py for the define).

Standard derivations applied (each preserves source data values):

* ****DM.AGE**** derived numeric from the de-identified `AGEGRP` (the PDS release masked exact age into `AGEGRP`;

subjects coded `>=85` are floored to `AGE=85`, with the cap flagged in `SUPPDM` `QNAM=AGEGRP`). The non-standard `AGEGRP` is removed from DM. `ACTARM`/`ACTARMCD` added (single completed arm, code `A`).

* **AE.AESOC** populated equal to `AEBODSYS` (MedDRA SOC already carried in `AEBODSYS`).

* **EPOCH** derived for AE/EX/VS (DS already carried it). Subject Elements (SE) are absent from the de-identified extract, so EPOCH is derived from the collected `VISIT` structure: `SCREENING`?`SCREENING`; `BASELINE`/`CYCLE n`/`END OF TREATMENT`?`TREATMENT`; `FOLLOW-UP n`?`FOLLOW-UP`; `UNSCHEDULED`?`TREATMENT`. All values are valid EPOCH CT.

* **EX.EXENDY** derived (study day of `EXENDTC` vs `RFSTDTC`).

* **Week-offset timing** (`AESTWK`/`AEENWK`/`AESTWKF`/`AEENWKF`, `DSSTWK`/`DSSTWKF`) -- non-standard in the parent domains -- relocated to **SUPPAE`/`SUPPDS** as supplemental qualifiers (linked by `IDVAR`/`IDVARVAL`), preserving the ± 3.5 -day timing (see §2 Date Precision Note and the date-precision sensitivity analysis, `06\_qc\_evidence/audit/run\_records/DATE\_PRECISION\_SENSITIVITY\_2026-06-20.md`).

* Redundant **SUBJID** removed from non-DM domains; non-standard `ARM2`/`ARMA`/`ARMCD2` (define-only phantoms, never in data) dropped from `IG.DM`.

* **TS** enriched with public NCT00417079 parameters (`NARMS=2`, `ACTSUB=371`, `SSTDTC=2007`, `AGEMIN=P18Y`). **TA** (Trial Arms) built from the public two-arm design.

* Variable labels title-cased, leading/trailing whitespace stripped, variable order aligned to the CDISC SDTM library.

5.1 CORE SDTMIG 3.4 -- honesty (F-015)

Run record: platform/conformance/CORE_SDTM34_RUN_RECORD.md

Engine: cdisc-rules-engine (CORE) v0.16.0 standard SDTMIG 3.4 CT 2026-03-27

Headline: targeted structural uplift rules cleared; overall issue count is not zero and must not be marketed as "CORE clean."

Class	Examples	Disposition (Path A)
Structural targets fixed	AESOC, AGE, EPOCH, EXENDY, non-standard?SUPP, labels/order/type	Accept as uplift success
Inherent de-identification	SITEID / COUNTRY / MedDRA hierarchy codes / exact AE dates removed by PDS	Accept -- cannot invent PII or lost codes
Real source-data quality	AESER consistency (CORE-000266/022); VSSTRESC/N (CORE-000732)	Accept -- do not overwrite true safety source to greenwash
Cross-domain N/A	RELREC/FAOBJ (CORE-000767) without FA in analysis-scoped package	Waive / N/A for this package scope
Engine-internal	CORE-000929 / CORE-001081 evaluation dataset failed to build	Accept with note -- tool noise, not silent data fix

Rule for reviewers:

* **Do claim:** "Uplifted package layer; CORE run recorded; structural targets cleared; residuals classified."

* **Do not claim:** "Full commercial conformance" or "zero CORE findings."

* **Rule-level disposition matrix (WS-1):**
[`docs/workstreams/WS1\_CORE\_RESIDUAL\_MATRIX.csv`](../../docs/workstreams/WS1_CORE_RESIDUAL_MATRIX.csv) -- filed; still not a "CORE clean" claim.

Related residual: week-offset / partial ISO dates -- F-017 (§2 AE note, §4.5). Never invent day precision.

Pinnacle 21 commercial ADaM/SDTM: NOT_AVAILABLE under Path A (see WS-3 external validation index). Local CORE + dual-language ADaM recon are substitutes only.